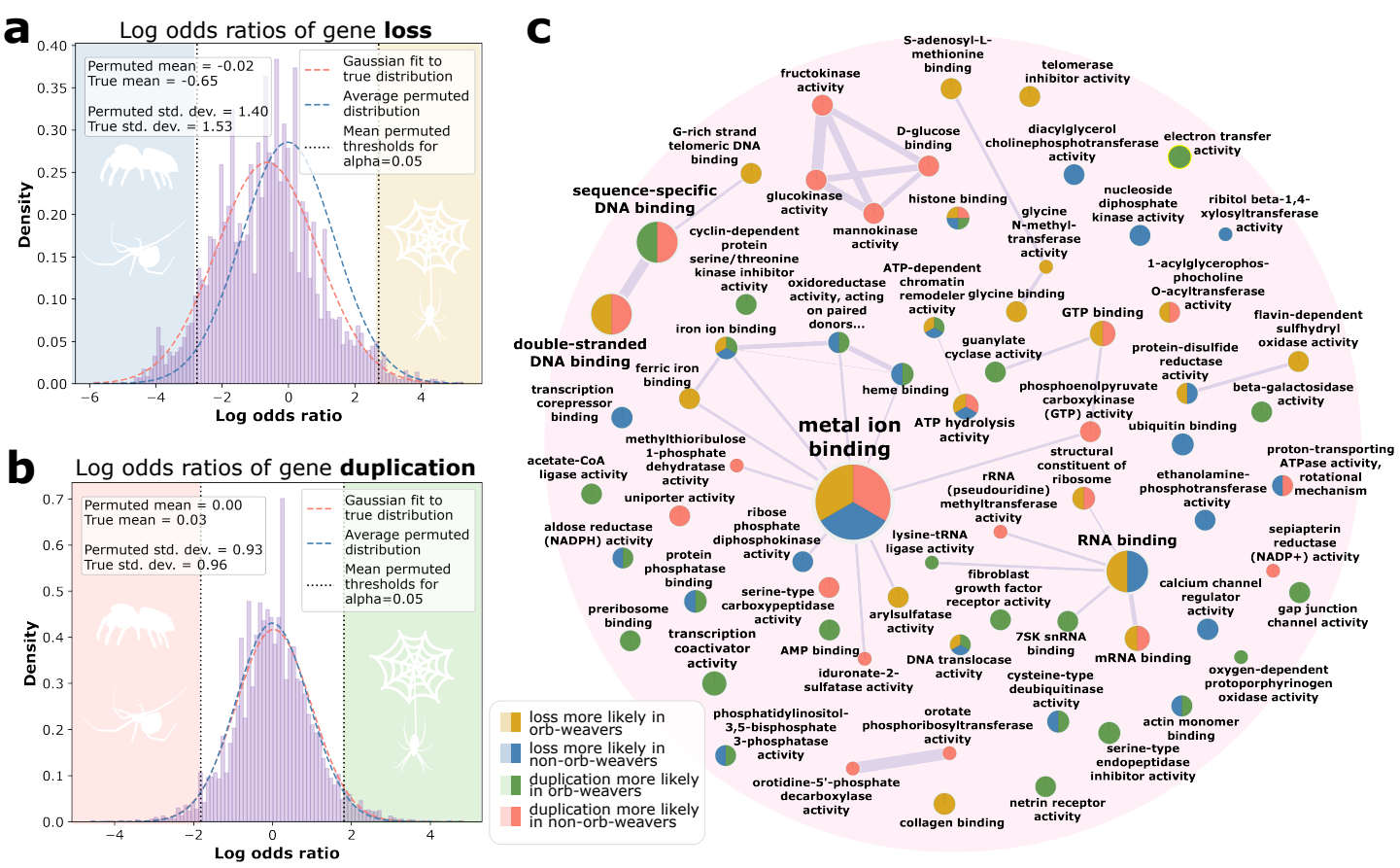

a
Log odds ratios of gene loss
Permuted mean = -0.02
True mean = -0.65
Permuted std. dev. = 1.40
True std. dev. = 1.53
Gaussian fit to true distribution
Average permuted distribution
Mean permuted thresholds for alpha=0.05
Density
Log odds ratio
b
Log odds ratios of gene duplication
Permuted mean = 0.00
True mean = 0.03
Permuted std. dev. = 0.93
True std. dev. = 0.96
Gaussian fit to true distribution
Average permuted distribution
Mean permuted thresholds for alpha=0.05
Density
Log odds ratio
c
S-adenosyl-L-methionine binding
telomerase inhibitor activity
fructokinase activity
G-rich strand telomeric DNA binding
D-glucose binding
diacylglycerol cholinephosphotransferase activity
electron transfer activity
sequence-specific DNA binding
glucokinase activity
histone binding
nucleoside diphosphate kinase activity
ribitol beta-1,4-xylosyltransferase activity
cyclin-dependent protein serine/threonine kinase inhibitor activity
mannokinase activity
glycine N-methyl-transferase activity
ATP-dependent chromatin remodeler activity
glycine binding
1-acylglycerophosphocholine O-acyltransferase activity
oxidoreductase activity, acting on paired donors...
iron ion binding
GTP binding
flavin-dependent sulfhydryl oxidase activity
double-stranded DNA binding
guanylate cyclase activity
protein-disulfide reductase activity
ferric iron binding
heme binding
phosphoenolpyruvate carboxykinase (GTP) activity
beta-galactosidase activity
transcription corepressor binding
ATP hydrolysis activity
metal ion binding
ubiquitin binding
methylthioribulose 1-phosphate dehydratase activity
structural constituent of ribosome
proton-transporting ATPase activity, rotational mechanism
acetate-CoA ligase activity
rRNA (pseudouridine) methyltransferase activity
ethanolamine-phosphotransferase activity
uniporter activity
aldose reductase (NADPH) activity
ribose phosphate diphosphokinase activity
lysine-tRNA ligase activity
RNA binding
sepiapterin reductase (NADP+) activity
protein phosphatase binding
fibroblast growth factor receptor activity
calcium channel regulator activity
serine-type carboxypeptidase activity
arylsulfatase activity
gap junction channel activity
preribosome binding
7SK snRNA binding
transcription coactivator activity
AMP binding
DNA translocase activity
mRNA binding
oxygen-dependent protoporphyrinogen oxidase activity
iduronate-2-sulfatase activity
cysteine-type deubiquitinase activity
phosphatidylinositol-3,5-bisphosphate 3-phosphatase activity
orotate phosphoribosyltransferase activity
actin monomer binding
serine-type endopeptidase inhibitor activity
orotidine-5'-phosphate decarboxylase activity
netrin receptor activity
collagen binding
loss more likely in orb-weavers
loss more likely in non-orb-weavers
duplication more likely in orb-weavers
duplication more likely in non-orb-weavers